

STT 861: Theory of Probability & Statistics I

Lecture 6: Bayes Theorem & Independence

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Sep 17, 2026 — Wells Hall C506

Reference: Casella & Berger Ch. 1.3

Lecture 6 Outline

Today

- Bayes' Theorem
- Independence of two and multiple events
- Theorem 1.3.9 and implications

Bayes' Theorem

Bayes' Theorem

If $A_1, \dots, A_k$ partition S , $P(A_i) > 0$, $P(B) > 0$:

$$P(A_j | B) = \frac{P(B | A_j) P(A_j)}{\sum_{i=1}^k P(B | A_i) P(A_i)}$$

$P(A_j)$ = **prior**; $P(A_j | B)$ = **posterior**.

Medical Diagnosis

$P(\text{disease}) = 0.01$, $P(+ | \text{disease}) = 0.95$, $P(+ | \text{no disease}) = 0.10$.

$$P(\text{disease} | +) = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.10 \times 0.99} = \frac{0.0095}{0.1085} \approx \mathbf{8.8\%}$$

Even with a positive test, only $\sim 9\%$ chance of having the disease!

Bayes' Theorem – Interpretation

Prior, Likelihood, Posterior

H = hypothesis, D = data (observed event).

$P(H)$ = **prior probability** (belief before seeing data)

$P(D|H)$ = **likelihood** (probability of data given H is true)

$P(H|D)$ = **posterior probability** (updated belief after data)

Bayes: $P(H|D) = \frac{P(D|H)P(H)}{P(D)}$ (“update prior with likelihood”)

Breast Cancer Screening – Bayes in Action

CB Example: Mammography Screening

Women aged 40–65: prevalence $P(C) = 0.00167$.

Sensitivity: $P(+|C) = 0.90$. Specificity: $P(-|\bar{C}) = 0.99$.

$$\begin{aligned} P(C|+) &= \frac{P(+|C)P(C)}{P(+|C)P(C) + P(+|\bar{C})P(\bar{C})} = \frac{0.90 \times 0.00167}{0.90 \times 0.00167 + 0.01 \times 0.99833} \\ &= \frac{0.001503}{0.001503 + 0.009983} = \frac{0.001503}{0.011486} \approx \mathbf{0.131} \end{aligned}$$

Despite positive test: only $\sim 13\%$ probability of cancer.

If negative: $P(C|-) = P(C|+, \text{neg test}) \approx 0.00017$.

Example: Lung Cancer and Cigarette Smoking

Study Design

Subjects are randomly assigned to two groups:

- E = heavy daily smoker $\bar{E}$ = non-smoker
- D = lung cancer $\bar{D}$ = no lung cancer

Question: Are D and E **independent**?

Independence

Definition

Events A and B are **independent** (AB) if $P(A \cap B) = P(A)P(B)$.

Equivalently (when $P(B) > 0$): $P(A|B) = P(A)$.

Theorem 1.3.9

If AB , then also: AB^c , A^cB , A^cB^c .

Proof of AB^c : $P(AB^c) = P(A) - P(AB) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c)$.

□

Mutual Independence (3 events)

A, B, C are **mutually independent** iff $P(ABC) = P(A)P(B)P(C)$ and all three pairs are independent.

Caution: pairwise independence $\nRightarrow$ mutual independence!

Independence Check

D and E are independent if and only if

$$P(D | E) = P(D) = P(D | \bar{E})$$

Equivalently:

$$P(D | E) = P(D | \bar{E})$$

Knowing someone smokes gives *no additional information* about their probability of lung cancer.

Remark

In practice, $P(D | E) \gg P(D | \bar{E})$, indicating that smoking and lung cancer are **not** independent. The odds ratio quantifies *how far* from independence we are.

Odds of Disease With and Without Exposure

Odds

The **odds** of an event A is $\frac{P(A)}{P(A^c)}$.

Odds of disease given exposure:

$$\begin{aligned}O(D | E) &= \frac{P(D | E)}{P(\bar{D} | E)} \\&= \frac{P(E | D) P(D) / P(E)}{P(E | \bar{D}) P(\bar{D}) / P(E)} \\&= \frac{P(E | D)}{P(E | \bar{D})} \cdot \frac{P(D)}{P(\bar{D})}\end{aligned}$$

Odds of disease without exposure:

$$\begin{aligned}O(D | \bar{E}) &= \frac{P(D | \bar{E})}{P(\bar{D} | \bar{E})} \\&= \frac{P(\bar{E} | D)}{P(\bar{E} | \bar{D})} \cdot \frac{P(D)}{P(\bar{D})}\end{aligned}$$

Both expressions share the same factor $\frac{P(D)}{P(\bar{D})}$

(the **prior odds** of disease).

The Odds Ratio

Odds Ratio (OR)

The **odds ratio** of disease with versus without exposure is:

$$\text{OR} = \frac{O(D | E)}{O(D | \bar{E})} = \frac{P(E | D) / P(\bar{E} | D)}{P(E | \bar{D}) / P(\bar{E} | \bar{D})}$$

Key Symmetry

The odds ratio of **disease** with/without exposure equals the odds ratio of **exposure** with/without disease:

$$\frac{O(D | E)}{O(D | \bar{E})} = \frac{O(E | D)}{O(E | \bar{D})}$$

$$\text{where } O(E | D) = \frac{P(E | D)}{P(\bar{E} | D)} \text{ and } O(E | \bar{D}) = \frac{P(E | \bar{D})}{P(\bar{E} | \bar{D})}.$$

Odds Ratio = 1 $\iff$ Independence

Theorem

D and E are **independent** if and only if $OR = 1$.

Proof ($\Rightarrow$): If $D \perp\!\!\!\perp E$, then $P(D | E) = P(D)$ and $P(D | \bar{E}) = P(D)$, so

$$O(D | E) = O(D | \bar{E}) \implies OR = 1.$$

Proof ($\Leftarrow$): If $OR = 1$, then $O(E | D) = O(E | \bar{D})$, which gives $P(E | D) = P(E | \bar{D})$, which implies $P(D | E) = P(D | \bar{E})$, and hence $P(D | E) = P(D)$, i.e. $D \perp\!\!\!\perp E$. $\square$

Case-Control Study Application

In a case-control study we recruit samples of cases (D) and controls ($\bar{D}$) and record exposure status. This gives us estimates of $P(E | D)$ and $P(E | \bar{D})$, and hence

$$\widehat{\text{OR}} = \frac{P(E | D) / P(\bar{E} | D)}{P(E | \bar{D}) / P(\bar{E} | \bar{D})}$$

By the symmetry of the OR, this estimates the odds ratio of *disease* with/without exposure — the quantity of primary scientific interest.

Conditional Independence

Conditional Independence

A and B are **conditionally independent given** C if:

$$P(AB|C) = P(A|C)P(B|C)$$

Equivalently: $P(A|BC) = P(A|C)$ – knowing B adds no information once C is known.

Important Questions

- ❶ Does AB imply $AB|C$? **No.** (*Simpson's Paradox*)
- ❷ Does $AB|C$ imply AB ? **No.**

Example (Simpson's Paradox): A drug appears beneficial overall but harmful within each gender subgroup – due to confounding by $C=\text{gender}$.

Pairwise vs. Mutual Independence

Independence for Three Events (CB Def 1.3.7)

A, B, C are **mutually independent** iff all four conditions hold:

$P(AB) = P(A)P(B)$, $P(AC) = P(A)P(C)$, $P(BC) = P(B)P(C)$,
and $P(ABC) = P(A)P(B)P(C)$.

Pairwise independence alone is **not** sufficient for mutual independence!

CB Example 1.3.10

Toss 2 fair coins. $A = \{H \text{ on coin 1}\}$, $B = \{H \text{ on coin 2}\}$, $C = \{\text{same result}\}$.

$P(A) = P(B) = P(C) = 1/2$; $P(AB) = P(AC) = P(BC) = 1/4$: pairwise independent.

But $P(ABC) = P(\{HH\}) = 1/4 \neq 1/8 = P(A)P(B)P(C)$. **Not mutually independent.**

Example 1.3.10 — Tossing Two Dice (Setup)

Example 1.3.10: Tossing Two Dice

Toss two fair dice. $S = \{(i, j) : i, j \in \{1, \dots, 6\}\}$, $|S| = 36$.

Define the events:

$$A = \{\text{doubles appear}\} = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

$$B = \{\text{sum is between 7 and 10}\}$$

$$C = \{\text{sum is 2, 7, or 8}\}$$

Individual probabilities:

$$P(A) = \frac{6}{36} = \frac{1}{6}$$

$$P(B) = \frac{18}{36} = \frac{1}{2}$$

$$P(C) = \frac{12}{36} = \frac{1}{3}$$

Counting B : sums 7, 8, 9, 10:

Sum 7: (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) — 6

Sum 8: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) — 5

Sum 9: (3, 6), (4, 5), (5, 4), (6, 3) — 4

Sum 10: (4, 6), (5, 5), (6, 4) — 3

Total: $6 + 5 + 4 + 3 = 18$. ✓

Example 1.3.10 — Three-Way Product Holds, But...

Three-Way Intersection

$$A \cap B \cap C = \{\text{sum is 8, composed of doubles}\} = \{(4, 4)\}$$

$$P(A \cap B \cap C) = \frac{1}{36} = \frac{1}{6} \times \frac{1}{2} \times \frac{1}{3} = P(A)P(B)P(C) \quad \checkmark$$

$$P(B \cap C) = P(\text{sum is 7 or 8}) = \frac{11}{36} \neq P(B)P(C) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} \approx 0.167$$

Similarly, $P(A \cap B) \neq P(A)P(B)$.

Conclusion: $P(A \cap B \cap C) = P(A)P(B)P(C)$ is **not sufficient** to guarantee mutual independence. We also need all pairwise products to hold.

Lesson

Mutual independence requires *every* subcollection to satisfy the product rule — not just the full triple.

Example 1.3.11 — Letters: Pairwise $\nRightarrow$ Mutual

Example 1.3.11: Letters

S = all $3! = 6$ permutations of $\{a, b, c\}$ plus the 3 triples:

$$S = \{aaa, abc, acb, bbb, bca, bac, ccc, cba, cab\}, \quad P(\text{each}) = \frac{1}{9}$$

Define $A_i = \{i\text{-th place is occupied by } \mathbf{a}\}$, $i = 1, 2, 3$.

Individual and pairwise:

$$P(A_i) = \frac{3}{9} = \frac{1}{3} \text{ for } i = 1, 2, 3.$$

$$P(A_1 \cap A_2) = P(A_1 \cap A_3) = P(A_2 \cap A_3) = \frac{1}{9}$$

$$= P(A_i)P(A_j) = \frac{1}{9} \checkmark$$

$\therefore$ The A_i are **pairwise independent**.

But the triple fails:

$$A_1 \cap A_2 \cap A_3 = \{aaa\}$$

$$P(A_1 \cap A_2 \cap A_3) = \frac{1}{9}$$

$$P(A_1)P(A_2)P(A_3) = \frac{1}{27}$$

$$\frac{1}{9} \neq \frac{1}{27} \quad \times$$

$\therefore$ **Not** mutually independent.

Definition 1.3.12 — Mutual Independence

Definition 1.3.12 (CB): Mutual Independence

A collection of events $A_1, A_2, \dots, A_n$ are **mutually independent** if for *any* subcollection $A_{i_1}, A_{i_2}, \dots, A_{i_k}$:

$$P\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k P(A_{i_j})$$

What this requires for $n = 3$ events

For A_1, A_2, A_3 to be mutually independent, **all four** conditions must hold simultaneously:

$$P(A_1 \cap A_2) = P(A_1)P(A_2)$$

$$P(A_2 \cap A_3) = P(A_2)P(A_3)$$

$$P(A_1 \cap A_3) = P(A_1)P(A_3)$$

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$$

Examples 1.3.10 and 1.3.11 show that none of these three conditions alone, nor any two together, is sufficient.

Example 1.3.13 — Three Coin Tosses: Mutual Independence

Example 1.3.13: Three Coin Tosses

Toss a fair coin 3 times.

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}, \quad P(\text{each}) = \frac{1}{8}$$

Let $H_i = \{i\text{-th toss is a Head}\}$, $i = 1, 2, 3$.

From equation (1.3.9): $H_1 = \{HHH, HHT, HTH, HTT\}$.

Individual probabilities:

$$P(H_1) = P(H_2) = P(H_3) = \frac{1}{2}$$

Pairwise (e.g. $H_1 \cap H_2$):

$$H_1 \cap H_2 = \{HHH, HHT\}$$

$$P(H_1 \cap H_2) = \frac{2}{8} = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(H_1)P(H_2) \checkmark$$

The same holds for $H_1 \cap H_3$ and $H_2 \cap H_3$.

Triple:

$$H_1 \cap H_2 \cap H_3 = \{HHH\}$$

$$P(H_1 \cap H_2 \cap H_3) = \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \checkmark$$

Conclusion

All four conditions hold.

H_1, H_2, H_3 are **mutually independent**.

Each toss has no effect on any other toss.

Summary: Examples 1.3.10–1.3.13

Example	Setup	Pairwise indep.	Mutually indep.
1.3.10	Two dice; events A, B, C $(P(A \cap B \cap C) = P(A)P(B)P(C))$ holds, but pairwise fails)	No	No
1.3.11	Letters; events A_1, A_2, A_3 (all pairs OK, but $P(A_1 \cap A_2 \cap A_3) \neq P(A_1)P(A_2)P(A_3)$)	Yes	No
1.3.13	Three coin tosses; events H_1, H_2, H_3	Yes	Yes

Take-away

Neither condition alone implies the other:

- $P(A_1 \cdots A_n) = P(A_1) \cdots P(A_n)$ **does not** imply pairwise independence.
- Pairwise independence **does not** imply mutual independence.
- **Mutual independence** (Def. 1.3.12) requires *all* subcollections.

Activity 6 – Bayes + Independence

Activity 6

- Ⓐ (Screening test) $P(\text{HIV}) = 0.002$, sensitivity=0.99, specificity=0.95.
Find $P(\text{HIV} \mid +)$ and $P(\text{HIV} \mid -)$.
- Ⓑ Roll a fair die. $A = \{2, 4, 6\}$, $B = \{1, 2, 3, 4\}$, $C = \{1, 3, 5\}$. Are A and B independent? Are A and C independent?
- Ⓒ With replacement: draw 2 cards. Are events “1st is $\diamond$ ” and “2nd is $\heartsuit$ ” independent?

Activity 6 – Solutions

- (a) $P(+) = 0.99(0.002) + 0.05(0.998) = 0.00198 + 0.0499 = 0.05188$
 $P(\text{HIV} | +) = 0.00198 / 0.05188 \approx \mathbf{3.8\%}$;
 $P(\text{HIV} | -) = P(\text{HIV}, -) / P(-) = 0.00002 / 0.94812 \approx \mathbf{0.002\%}$
- (b) $P(A) = 1/2$, $P(B) = 2/3$, $P(AB) = P(\{2, 4\}) = 2/6 = 1/3 = P(A)P(B)$. **Independent.**
 $P(C) = 1/2$, $P(AC) = P(\emptyset) = 0 \neq 1/4$. **Not independent.**
- (c) $P(1\text{st} \diamond) = 13/52 = 1/4$, $P(2\text{nd} \heartsuit) = 13/52 = 1/4$.
With replacement: draws are independent, $P(\text{both}) = 1/16 = 1/4 \times 1/4$. **Independent.**